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AI-driven weather forecasts for climate adaptation in India

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Advanced monsoon onset prediction with multi-week lead time via an artificial intelligence (AI) weather model helps smallholder farmers adapt to a changing climate.

Earlier in 2025, 38 million farmers in India received a first-of-its-kind weather forecast about the monsoon onset four weeks in advance. Led by the Indian Ministry of Agriculture and Farmers' Welfare in collaboration with the University of Chicago's Human-Centered Weather Forecasts (HCF) Initiative, a multi-disciplinary team of climate scientists, artificial intelligence (AI) experts, economists and social scientists rolled out an AI-based weather forecast programme to provide long-range forecasts about the monsoon arrival across India. This period of continuous rainfall marks the beginning of the rainy season in India, which is a crucial time for growing both staples and cash crops for more than 100 million farmers across the country.

Farmers usually rely on past knowledge, word-of-mouth updates, and government bulletins and advisories to anticipate when the rainfall might reach their specific region once onset has been declared. However, predicting the Indian monsoon has been a challenge for many decades. "Monsoons are notoriously complex continent-sized patterns of wind that persist for months and yet interact with large numbers of small-scale clouds that each last only a few hours," explains William Boos, an atmospheric scientist and monsoon expert at UC Berkeley, and a collaborator on the project.

Monsoon onset was identified as a crucial factor for Indian farmers by another team of researchers at the University of Chicago. They found that providing a monsoon forecast a month in advance resulted in substantial savings and losses averted for farmers in the Indian state of Telangana, who tailored their investments for planting based on this information. The results from the experimental study substantiated that long-range forecasts can help farmers adapt to climate change[1].

However, running physics-based numerical weather prediction models requires a lot of computing power, a highly skilled workforce to make them work for specific use cases, and most models provide reliable forecasts only a few days in advance. In this respect, recent advancements in AI weather forecasting have been a game-changer.

"This is really a revolution that started in early 2022 when two papers[2,3] showed that a deep neural net trained on ERA5 can have forecast accuracy comparable to the best numerical weather models," says Pedram Hassanzadeh, an AI expert and extreme weather researcher at the University of Chicago. "And once trained, these can be 100,000 times faster to run [than numerical weather predictions]. You could run it on a laptop and do the best weather forecast."

In order to generate monsoon forecasts for India, Google's NeuralGCM and the European Centre for Medium-Range Weather Forecasts' Artificial Intelligence Forecasting System and Integrated Forecasting System were chosen.

Mathematically blending these systems with 125 years of historical rainfall data over India, a probabilistic model with up to a 30-day lead time was produced, resulting in a calibrated probabilistic forecast that showed better skill than either of the individual models, says Mayank Gupta, a researcher who is part of the HCF initiative and a lead at the Development Innovation Lab India, an organization that works closely with the Indian Government alongside other researchers and institutions to test and scale interventions in the country.

Months before the monsoon season, a team based in India had been conducting iterative field testing until the farmers' comprehension of the probabilistic forecast messages improved. "You can have the best forecasts, but if the farmer doesn't understand it, that is a policy failure," says Zoya Khan, a policy expert at the Development Innovation Lab India, who led the efforts to understand what sort of messages worked best for farmers and to facilitate dissemination.

In the run-up to the monsoon, weekly onset forecast messages were sent to farmers across 13 states in 5 regional languages via SMS through a government service portal (mKisan), which provides mobile-based advisories to registered farmers. The blended AI model accurately predicted this year's early monsoon arrival and also forecasted an unusual dry spell, where the monsoon progression stalled for approximately three weeks before resuming its northward movement. The current model forecasts at a resolution of a 200-km grid, providing farmers across many parts of the country with localized rainfall updates like never before.

The current AI weather models account for the changes in climate that have happened so far and are expected to be updated routinely with the most recent meteorological data, says Hassanzadeh. "They provide a critical climate adaptation tool, as high-quality forecasts can help farmers and others to better respond to weather shocks."

Monitoring surveys also show a near universal demand for these services, says Khan. "Ninety seven per cent of farmers said that they would like to receive such information in future."

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Competing interests

The author declares no competing interests.

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